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**Beyond GARCH? A Rigorous Comparison of Classical and Deep Learning
Models for Financial Volatility Forecasting**

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Abstract

Forecasts of financial volatility underpin risk management, derivative pricing and portfolio construction. A substantial recent literature reports that deep learning models outperform the classical econometric models long standard in the field, yet this sits uneasily beside earlier evidence that well-specified classical models are difficult to surpass. Many of the comparisons involved evaluate forecasts against noisy proxies for an unobservable target, specify their baselines without verification, rely on a single fixed data split, and omit tests of statistical significance, so it is unclear whether the reported improvements are real.

This dissertation subjects the claim to a deliberately rigorous test. Six models are compared on identical data: an exponentially weighted baseline, GARCH(1,1), two asymmetric GARCH specifications, a heterogeneous autoregressive model, a long short-term memory network, and a GARCH-LSTM hybrid. All are evaluated on 4,005 one-day-ahead forecasts of S&P 500 realised volatility from 2004 to 2022, using an expanding-origin walk-forward protocol in which look-ahead bias is prevented by construction and verified by adversarial tests. Accuracy is measured with the QLIKE loss, differences are assessed with the Diebold-Mariano test, results are disaggregated across calm and crisis regimes, and the classical baselines are validated against published parameter estimates for the same index.

Neither the network nor the hybrid significantly outperforms the heterogeneous autoregressive model, a linear regression on three lagged averages of realised volatility. Two ablation studies explain why. Deprived of an exogenous implied-volatility feature, and thus reduced to the information available to the classical benchmark, the network falls significantly behind it: on equal information the deep architecture is a liability rather than an advantage, and the network's apparent parity derives entirely from the additional information it receives. The econometric structural feature has no aggregate effect but redistributes forecast skill toward crisis periods.

The study contributes a decomposition of forecast skill into architecture, information and econometric structure, and demonstrates that a reported deep learning advantage in this domain may reflect the data supplied to the model rather than the model itself. The deep models further consumed computation orders of magnitude greater than the classical benchmark for accuracy statistically indistinguishable from it.

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1 Introduction

Financial markets are characterised less by the predictability of their direction than by the predictability of their turbulence. While decades of empirical research suggest that the direction of asset price movements is close to unforecastable, the magnitude of those movements is demonstrably not. Periods of market calm cluster together, as do periods of crisis. This asymmetry, of unpredictable returns but predictable variance, is the foundation upon which the entire discipline of volatility forecasting rests, and it is the starting point of this dissertation.

1.1 Background

Volatility, in the financial context, refers to the dispersion of asset returns over time. It is conventionally measured as the standard deviation, or equivalently the variance, of returns over a given horizon. Its importance is difficult to overstate: volatility is a direct measure of risk, and consequently sits at the centre of nearly every quantitative decision made in modern finance. Value-at-Risk (VaR) calculations, which determine the capital that financial institutions must hold against potential losses, depend on forecasts of future volatility. Option pricing models take volatility as a primary input. Portfolio construction, position sizing, and margin requirements are all calibrated against expected volatility. A forecast of volatility is, in effect, a forecast of risk itself.

The empirical behaviour of volatility exhibits several well-documented regularities. The most prominent of these is volatility clustering: large price movements tend to be followed by further large movements, and small movements by further small movements, irrespective of direction (Mandelbrot, 1963; Cont, 2001). Statistically, this manifests as a striking asymmetry in the autocorrelation structure of returns. The autocorrelation of returns themselves is approximately zero, consistent with the efficient markets hypothesis, while the autocorrelation of squared returns is significantly positive and decays only slowly. It is precisely this asymmetry that makes volatility forecasting a tractable problem where return forecasting largely is not.

Alongside clustering, financial returns display fat tails, meaning that extreme movements occur far more frequently than a Gaussian distribution would predict, and a pronounced leverage effect, whereby negative returns tend to raise subsequent volatility by more than positive returns of comparable magnitude (Black, 1976). Volatility is also persistent yet mean-reverting: shocks decay gradually, and volatility drifts back toward a long-run average rather than diverging. Any model that aspires to forecast volatility adequately must, in some form, accommodate these four features.

There is, however, a methodological complication that distinguishes volatility forecasting from most other forecasting problems, and it is one that shapes this entire dissertation. Volatility is a latent quantity: it is never directly observed. On any given trading day the market reveals a single return, not that day's variance. A volatility forecast can therefore never be compared against the truth. It can only be compared against a proxy for the truth. The consequence is that the quality of any evaluation is bounded above by the quality of the proxy chosen, and, as Chapter 2 will demonstrate, an inappropriate choice of proxy has historically been sufficient to invert the conclusions of an entire literature (Andersen & Bollerslev, 1998). This dissertation therefore treats the construction of the forecasting target as a first-order methodological decision rather than an implementation detail.

The dominant approach to volatility modelling emerged from econometrics. The ARCH framework of Engle (1982) and its generalisation to GARCH by Bollerslev (1986) model conditional variance as an explicit function of past shocks and past variance, encoding clustering and mean reversion directly into the model structure. These models remain the field standard some four decades later. More recently, the availability of deep learning architectures capable of learning arbitrary nonlinear temporal dependencies has prompted a wave of research applying recurrent and attention-based networks to volatility forecasting. The relative merits of these two traditions form the subject of this study.

1.2 Motivation

The motivation for this dissertation arises from a question that the existing literature leaves unresolved.

A substantial body of econometric evidence suggests that classical volatility models are remarkably difficult to improve upon. In perhaps the most comprehensive comparison of its kind, Hansen and Lunde (2005) evaluated 330 volatility model specifications and found no convincing evidence that more elaborate models outperformed a simple GARCH(1,1) for exchange-rate data, although models incorporating a leverage effect were found to be superior for equity returns. This finding is less surprising than it first appears. A GARCH(1,1) model captures volatility clustering, persistence, and mean reversion using approximately three parameters. It has almost nothing to overfit, and it does not need to learn the structure of volatility from data because that structure is imposed a priori by economic reasoning. Parsimony, in this setting, is a genuine strength. It is important to note, however, that this evidence predates the widespread adoption of deep learning, which gathered pace only from around 2012.

More recent machine learning research reports that deep sequence models, and long short-term memory networks in particular, outperform GARCH-family baselines,

sometimes substantially (Bucci, 2020; Kim & Won, 2018). Neural networks impose no functional form and can, in principle, capture nonlinear dependencies that a fixed GARCH recursion cannot express. They can also ingest exogenous information such as implied volatility, trading volume and macroeconomic variables, for which the GARCH framework has no native mechanism.

These two bodies of evidence are separated by roughly a decade of progress in machine learning and are therefore not straightforwardly contradictory: it is quite possible that GARCH was the stronger approach in 2005 and that deep models have since closed or reversed the gap. The open question is whether the more recent claims of deep learning superiority survive rigorous evaluation. A close reading of the comparative literature suggests grounds for caution. Many studies evaluate forecasts against squared daily returns, a proxy so noisy that genuine differences between models may be obscured entirely. Baseline GARCH specifications are frequently reported without asymmetric terms or estimation diagnostics, making it impossible to verify whether the benchmark was specified fairly, and beating a mis specified baseline establishes nothing. Evaluation is commonly conducted on a single fixed train-test split rather than across multiple forecast origins. Tests of statistical significance are rarely applied, so it cannot be determined whether a reported margin exceeds sampling variation. The seed-to-seed variability of deep models, which is known to be considerable, is seldom reported at all.

Taken together, these observations suggest that whether deep learning genuinely improves volatility forecasting remains an open question, not because the models have not been compared, but because the comparisons have not consistently been conducted under conditions rigorous enough for their conclusions to be trusted. The motivation for this dissertation is therefore not to propose a new architecture, but to subject the more recent claims to a fair test.

A secondary motivation follows from the first. Where deep models do outperform classical baselines, the literature rarely establishes why. If a neural network is supplied with implied volatility data that GARCH structurally cannot use, any resulting improvement may be attributable to superior information rather than to superior architecture. These are very different claims with very different implications, and distinguishing between them requires an ablation study of the kind this dissertation undertakes.

1.3 Aims and Objectives

The aim of this dissertation is to determine whether deep learning models improve the accuracy of realised volatility forecasts relative to classical GARCH-family baselines, under a leakage-free evaluation protocol employing robust proper scoring rules and formal tests of statistical significance; and, where improvements are

observed, to identify whether they derive from model architecture or from access to exogenous information.

The study is conducted on the S&P 500, accessed via the SPY exchange-traded fund, which is chosen for the length and quality of its available history and because its extensive treatment in the existing literature permits the classical baselines to be validated against published parameter estimates.

To achieve this aim, the following objectives are defined:

- **O1: Literature Review.** Critically review the literature on classical volatility modelling, deep learning approaches to volatility forecasting, and hybrid architectures; and, in particular, examine the methodological adequacy of existing comparisons in order to identify a defensible research gap.
- **O2: Data Acquisition and Target Construction.** Acquire daily returns and high-frequency realised volatility data for the S&P 500, together with the VIX implied volatility index as an exogenous predictor; construct and validate a realised volatility forecasting target, justifying this choice over noisier alternatives.
- **O3: Evaluation Protocol.** Design and implement a walk-forward evaluation harness that is free of look-ahead bias by construction, incorporating the QLIKE robust loss function and the Diebold-Mariano test of equal predictive accuracy. This harness is fixed prior to any model being built.
- **O4: Model Ladder.** Implement a ladder of models of increasing complexity: an EWMA baseline; GARCH(1,1); the asymmetric EGARCH and GJR-GARCH specifications; the HAR-RV model; a long short-term memory (LSTM) network, a form of recurrent neural network suited to sequential data (introduced in Section 3.3); and a GARCH-LSTM hybrid in which the conditional variance forecast produced by GARCH is supplied to the network as an input feature. As part of this objective, the GARCH(1,1) baseline is validated against parameter estimates published for the S&P 500 in the existing literature, in order to confirm that the classical benchmark has been specified correctly before any comparison is drawn.
- **O5: Comparative Evaluation and Ablation.** Evaluate all models on identical walk-forward splits; test the statistical significance of pairwise differences; disaggregate performance by market regime; report stability across random seeds; and conduct an ablation study removing the exogenous VIX feature and the GARCH-derived feature in order to attribute any observed gains to their source.

The ordering of these objectives is deliberate. O3 precedes O4 because designing the evaluation protocol before implementing any model removes the possibility,

however unintentional, of specifying an evaluation that flatters a favoured architecture. This inversion of the conventional order is itself a methodological commitment, and it is defended in Chapter 3.

1.4 Structure of the Report

The remainder of this dissertation is organised as follows.

- **Chapter 2: Literature Review.** Establishes the stylised facts of financial returns; examines the latent nature of volatility and the consequences of proxy choice; traces the development of classical volatility models from ARCH through to HAR-RV; surveys deep learning and hybrid approaches; reviews the methodology of forecast evaluation; and concludes by identifying the methodological gap that this study addresses.
- **Chapter 3: Methodology.** Analyses the principal threats to the validity of a volatility forecasting comparison, namely look-ahead bias and data leakage, the unobservable target, baseline fairness, overfitting under limited data, and regime dependence, and sets out the design choices adopted to mitigate each, with reference to the literature.
- **Chapter 4: Implementation.** Documents the data sources and their provenance, the construction and validation of the realised volatility target, the feature set, the exploratory analysis, the evaluation harness, and the implementation of each model in the ladder.
- **Chapter 5: Presentation of Results.** Reports forecast accuracy across all models, the results of pairwise Diebold-Mariano significance testing, performance disaggregated by market regime, stability across random seeds, and the ablation study.
- **Chapter 6: Critical Review of Results.** Interprets the findings; addresses directly whether additional model complexity improves forecast accuracy; examines the contribution of exogenous information relative to architecture; discusses regime dependence; and sets out the threats to validity of the study.
- **Chapter 7: Critical Review of Project Objectives.** Assesses the extent to which each objective defined in Section 1.3 was met, including an honest account of any components descoped.
- **Chapter 8: Conclusion.** Summarises the findings and their implications, and identifies avenues for future work.

2 Literature Review

This chapter reviews the literature relevant to the comparison of classical and deep learning approaches to volatility forecasting. It is organised as a single argument rather than a survey. It begins with the empirical properties of financial returns that any volatility model must accommodate, then examines the problem that volatility cannot be directly observed and the consequences this has for measurement. It traces the development of classical volatility models and then the deep learning approaches that have more recently been applied to the same problem, before considering hybrid models that seek to combine them. It then reviews the methodology of forecast evaluation, which is central to this study, and concludes by identifying the methodological gap that the dissertation addresses.

2.1 Stylised Facts of Financial Returns

Financial return series display a number of empirical regularities that are remarkably consistent across assets, markets and time periods. These regularities, often termed stylised facts, are important here because each of the models discussed later in the chapter exists in order to capture one or more of them (Cont, 2001). Understanding them is a precondition for assessing whether any given model is well suited to the task.

The most consequential regularity is a striking asymmetry in predictability. The direction of returns is close to unforecastable, consistent with the efficient markets hypothesis, yet their magnitude is not. Statistically, this appears as a near-zero autocorrelation in returns themselves alongside a strong and slowly decaying positive autocorrelation in squared or absolute returns. It is this second property that makes volatility, unlike direction, a tractable forecasting target, and it is the empirical foundation on which the entire field rests.

Closely related is volatility clustering, the tendency for large price movements to be followed by further large movements and calm periods by further calm, irrespective of direction. First observed by Mandelbrot (1963) and since documented extensively, clustering is among the most robust regularities in empirical finance. It implies that volatility is persistent and can be forecast from its own recent history, which is precisely what the models in this chapter attempt to do.

Return distributions also exhibit fat tails, meaning that extreme movements occur far more frequently than a normal distribution would predict. This excess kurtosis has a direct methodological consequence: models that assume Gaussian innovations are mis specified, which is why heavier-tailed distributions such as the Student-t are commonly used when estimating volatility models. A further asymmetry, the leverage effect, describes the tendency of negative returns to raise subsequent volatility by

more than positive returns of comparable magnitude (Black, 1976). This asymmetry is not captured by the earliest volatility models and directly motivates the asymmetric extensions discussed in Section 2.3.

Finally, volatility is persistent but mean-reverting: shocks to volatility decay only gradually, yet volatility drifts back toward a long-run average rather than growing without bound. Taken together, these facts amount to a specification that any adequate volatility model must satisfy. It must capture clustering, persistence with mean reversion, heavy tails and asymmetry. This raises the question that organises the remainder of the chapter: is it more effective to build this structure into a model in advance, as econometrics does, or to allow a flexible model to learn it from data, as deep learning attempts? This is the tension the dissertation examines.

2.2 The Latent Target Problem

A feature that distinguishes volatility forecasting from most other forecasting problems, and one that shapes this entire study, is that volatility is not directly observable. On any given trading day the market reveals a single return, not the variance of the process that generated it. A volatility forecast can therefore never be compared against the truth, only against a proxy for the truth constructed from observed data. The consequence, easily overlooked, is that the quality of any evaluation is bounded above by the quality of the proxy chosen.

The most immediate proxy is the squared daily return, which is an unbiased estimator of the day's variance. It is, however, an extremely noisy one, since it is a single realisation of a random variable whose expectation is the quantity of interest. When forecasts are evaluated against so noisy a proxy, all models appear to forecast poorly, the reported explanatory power of volatility models is famously low, and genuine differences between competing models can be entirely obscured by the noise in the proxy itself.

The importance of this point is best illustrated by an episode in the development of the field. Through the 1990s, GARCH models were widely believed to produce poor volatility forecasts, a belief based on their evaluation against squared returns. Andersen and Bollerslev (1998) demonstrated that this apparent failure was an artefact of the measuring instrument rather than a property of the models. When forecasts were instead evaluated against realised volatility, a far more accurate proxy, standard volatility models were shown to forecast well. The significance of this result for the present study is difficult to overstate: it establishes that the choice of proxy alone can reverse the apparent conclusion of an entire literature, and it is the precedent for treating the construction of the forecasting target as a first-order methodological decision.

Realised volatility is constructed by summing squared intraday returns, for example at five-minute intervals, over a trading day. Because it aggregates many observations, it provides a much more precise estimate of the latent daily variance than a single squared daily return (Andersen et al., 2003). The choice of sampling frequency involves a trade-off, since sampling too finely introduces distortions from market microstructure, and the five-minute convention is a widely adopted compromise. Where intraday data is unavailable, range-based estimators that use the daily open, high, low and close, such as those of Garman and Klass (1980) and Yang and Zhang (2000), offer an intermediate improvement over squared returns without requiring high-frequency data.

The implication for this dissertation is that realised volatility is adopted as the forecasting target, and that any study evaluating volatility forecasts against squared returns must be read with corresponding caution. This caution informs the critical assessment of the comparative literature in Section 2.7.

2.3 Classical Volatility Models

The dominant approach to volatility modelling originates in econometrics, and its defining characteristic is that the structure of volatility is imposed on the model in advance rather than learned from data. Because the classical models constitute the baseline against which the deep learning approaches are assessed, it is important to present them not as a formality but as the strong and well-founded benchmark they are.

2.3.1 From ARCH to GARCH

The foundational insight is due to Engle (1982), whose ARCH model treats the variance of returns as time-varying and conditional on the recent history of the series, expressing the current variance as a function of recent squared shocks. This directly encodes volatility clustering. Bollerslev (1986) generalised the model to GARCH by allowing the current conditional variance to depend also on its own recent values, so that variance is a function of both the previous period's shock and the previous period's variance. In the widely used GARCH(1,1) specification, the parameters carry clear economic meaning: one governs the reaction of volatility to new information, another its persistence, and their sum determines how quickly the effect of a shock decays.

The parsimony of this specification is a genuine strength rather than a limitation, and it is central to the comparison that follows. A GARCH(1,1) model captures clustering, persistence and mean reversion using approximately three parameters. There is very little for it to overfit, and it does not need to learn the structure of volatility from data because that structure is supplied a priori by economic reasoning. This stands

in direct contrast to a deep network, which must infer the same structure from a limited and noisy sample using a far larger number of parameters. The classical model is, in effect, told what a neural network must discover.

2.3.2 Asymmetric extensions

The basic GARCH model treats positive and negative shocks symmetrically and therefore cannot capture the leverage effect described in Section 2.1. Two extensions address this. The EGARCH model of Nelson (1991) models the logarithm of the conditional variance, which guarantees positivity and allows the sign of a shock to influence volatility differently from its magnitude. The GJR-GARCH model of Glosten, Jagannathan and Runkle (1993) adds a term that activates only for negative shocks, directly capturing the greater impact of bad news. These models exist specifically to accommodate the asymmetry documented in Section 2.1, and their relevance to the present study is heightened by evidence, discussed below, that such asymmetry matters particularly for equity data.

2.3.3 The HAR-RV model

A more recent and distinct classical approach is the heterogeneous autoregressive model of realised volatility, or HAR-RV, proposed by Corsi (2009). Rather than modelling conditional variance from returns, it models realised volatility directly as a linear regression on its own averages over daily, weekly and monthly horizons. Its rationale is that market participants operate over different time horizons, and despite its simplicity it is a notably strong and difficult benchmark to surpass for realised volatility forecasting. Its inclusion here provides a strong classical competitor defined directly in terms of the realised volatility target used throughout the study.

2.3.4 Does anything beat a well-specified classical model?

The strength of the classical baseline is established most directly by Hansen and Lunde (2005), who compared 330 volatility model specifications. Their finding is more nuanced than it is sometimes reported. For exchange-rate data they found no convincing evidence that any model outperformed a simple GARCH(1,1); for equity returns, however, they found that models incorporating a leverage effect were clearly superior to the symmetric GARCH(1,1). Two implications follow for the present study, which uses equity data. First, the asymmetric specifications of Section 2.3.2 are not optional but necessary, and a symmetric GARCH(1,1) alone would be an inadequate baseline. Second, and more broadly, this body of evidence sets a demanding standard for any challenger and predates the modern deep learning methods examined next, which is precisely what makes the comparison an open question rather than a settled one.

2.4 Deep Learning Approaches

An alternative tradition, developed largely outside econometrics, imposes no functional form on the relationship being modelled and instead learns it from data. This flexibility is the counterpart to the structure of the classical models, and it defines the trade-off at the centre of the dissertation.

The most widely applied architectures for sequential data of this kind are recurrent neural networks, and in particular the long short-term memory network of Hochreiter and Schmidhuber (1997). Its gating mechanism allows information to be retained across long sequences and mitigates the vanishing-gradient problem that limits simpler recurrent networks, making it capable in principle of representing the persistence and clustering seen in volatility. Convolutional architectures adapted for sequences, such as temporal convolutional networks, and attention-based architectures including the temporal fusion transformer of Lim et al. (2021), offer alternative means of modelling temporal dependence, the latter being designed specifically for multi-horizon forecasting with exogenous inputs.

The most consequential difference between these models and the classical family is not architectural sophistication but access to information. A GARCH model conditions only on the past of the return series itself. A neural network can additionally ingest exogenous variables, such as implied volatility, trading volume or macroeconomic indicators, for which the GARCH framework has no native mechanism. This suggests that where a deep model does outperform a classical baseline, the improvement may derive from the information it is given rather than from the flexibility of its architecture. Distinguishing between these two sources of advantage is a central aim of this study and motivates the ablation described in later chapters.

These advantages carry corresponding costs. A deep network has a large number of parameters and is estimated from a limited and noisy sample containing few distinct volatility regimes, which creates a substantial risk of overfitting. Its performance can also vary appreciably across training runs owing to random initialisation, so that a single reported result may not be representative. It is computationally far more demanding than a GARCH model and offers little transparency into how its forecasts are formed. These considerations motivate the stability and evaluation safeguards adopted later in the study.

The empirical evidence on whether deep models outperform classical baselines is, on balance, mixed. Some studies report clear improvements (Bucci, 2020; Kim & Won, 2018), while others find little difference or favour the classical approach. A plausible explanation, developed in Section 2.7, is that much of this variation reflects differences in evaluation methodology rather than genuine differences in model

capability. A cautionary parallel is found in the adjacent literature on tabular data, where careful comparisons have repeatedly found that well-tuned classical methods match or exceed deep architectures (Shwartz-Ziv & Armon, 2022), suggesting that claims of deep learning superiority in structured-data domains warrant particularly careful scrutiny.

2.5 Hybrid Models

If the strength of the classical models lies in their encoded structure and the strength of the neural networks in their flexibility and capacity for exogenous inputs, a natural response is not to choose between them but to combine them. The common approach supplies the conditional variance estimated by a GARCH model as an input feature to a neural network, allowing the classical model to contribute the structure it captures well while the network models any remaining nonlinearity.

The canonical example is the work of Kim and Won (2018), who integrate the outputs of several GARCH-type models as inputs to an LSTM and report that the resulting hybrid outperforms both its constituents. The result is of direct relevance to this study, which includes a comparable hybrid. It is, however, instructive to examine the evaluation on which such conclusions rest. Where a hybrid is assessed on a single fixed split and without a test of statistical significance, it cannot be established that the reported advantage exceeds what might arise from sampling variation. This observation motivates the present study's use of significance testing across multiple forecast origins.

A further limitation of the hybrid literature is that it typically reports that a hybrid performs well without isolating why. If a hybrid combines GARCH structure with exogenous inputs and a flexible architecture, its advantage could originate in any of these. The ablation conducted in this study, which removes the GARCH-derived feature and the exogenous input in turn, is designed to attribute any observed gain to its source, addressing a gap that the existing hybrid literature largely leaves open.

2.6 Evaluating Volatility Forecasts

Because the comparison at the centre of this dissertation depends on measurement, the methodology of forecast evaluation is as important as the models themselves. Three aspects are considered: the choice of loss function, the assessment of statistical significance, and the validation protocol.

2.6.1 Loss functions under an imperfect proxy

Since the forecasting target is a noisy proxy for latent volatility, the choice of loss function is not a neutral one. Patton (2011) demonstrates that many intuitively reasonable loss functions produce a different ranking of models when computed

against a noisy proxy than they would against true volatility. This is a serious result: it means that an otherwise careful study can reach the wrong conclusion purely through its choice of loss function. Patton identifies a class of loss functions that remain robust, in the sense of preserving the true ranking despite proxy noise, of which the squared error and the QLIKE loss are the principal members. QLIKE has the additional property of penalising under-prediction of volatility more heavily than over-prediction, which is the appropriate asymmetry in a risk-management context where underestimating risk is the more dangerous error. For these reasons QLIKE is adopted as the primary loss function in this study, with the root mean squared error reported alongside for comparability with prior work.

2.6.2 Statistical significance of forecast differences

A lower average loss for one model than another does not establish that it forecasts more accurately, because the difference may be attributable to sampling variation. The test of Diebold and Mariano (1995) provides a formal assessment of the null hypothesis of equal predictive accuracy between two forecasts. Its relevance to the present study is underlined by the observation that much of the comparative volatility literature reports differences in loss without testing their significance, so that the reported margins cannot be distinguished from noise. This study therefore applies the Diebold-Mariano test to every pairwise comparison. The model confidence set of Hansen, Lunde and Nason (2011) generalises this idea to the simultaneous comparison of many models and is noted as a possible extension.

2.6.3 Validation protocol and look-ahead bias

The final consideration is the protocol by which forecasts are generated and evaluated. The random partitioning of data standard in cross-sectional machine learning is invalid for time series, because it permits a model to be trained on observations occurring after those on which it is tested (Bergmeir & Benitez, 2012). The appropriate alternative is walk-forward evaluation, in which the model is repeatedly estimated on data up to a given point and used to forecast the next, with the origin advancing through the sample. Beyond the choice of protocol, care is required to prevent subtler forms of look-ahead bias, including the scaling of features using statistics computed over the full sample, the estimation of parameters on the entire series, and the selection of hyper-parameters on data overlapping the test period. Each of these is addressed explicitly in the methodology of Chapter 3.

2.7 Critical Synthesis and Research Gap

The preceding sections permit a synthesis. The literature offers two accounts that stand in apparent tension. A substantial econometric literature, exemplified by Hansen and Lunde (2005), finds well-specified classical models very difficult to surpass. A more recent machine learning literature reports that deep models

outperform them. As established in Chapter 1, these accounts are separated by a decade of methodological progress and are therefore not straightforwardly contradictory; the question is whether the more recent claims withstand rigorous evaluation.

There are grounds to doubt that they have consistently been subjected to such evaluation. Several recurring weaknesses can be identified in the comparative literature. First, many studies evaluate against squared daily returns, a proxy that Section 2.2 showed can invert conclusions. Second, GARCH baselines are frequently specified without asymmetric terms or reported diagnostics, so that the fairness of the benchmark cannot be verified, despite the finding of Hansen and Lunde (2005) that asymmetry is essential for equity data. Third, evaluation is often conducted on a single fixed split rather than across multiple forecast origins. Fourth, tests of statistical significance are rarely applied, so reported differences cannot be distinguished from sampling variation. Fifth, the seed-to-seed variability of deep models is seldom reported, despite its known magnitude. Sixth, results are typically aggregated across the whole sample, obscuring any dependence of model ranking on market conditions. Not every study exhibits all of these weaknesses, and some avoid them, but their prevalence means that the balance of reported evidence cannot be taken at face value.

The gap this dissertation addresses is therefore not the absence of another model architecture, of which there are many, but the absence of a comparison rigorous enough for its conclusion to be trusted. Such a comparison requires a realised volatility target, a classical baseline validated against published results and including the asymmetric terms appropriate to equity data, leakage-free walk-forward evaluation, a robust loss function, formal significance testing, the reporting of stability across seeds, and disaggregation by market regime. The contribution of this study is accordingly methodological rather than architectural.

A second and distinct contribution concerns attribution. The existing literature, particularly on hybrid models, reports that a given model performs well but rarely establishes why. By ablating the exogenous input and the GARCH-derived feature, this study seeks to determine whether any observed advantage of the deep and hybrid models derives from architecture or from access to information, a distinction the literature has largely left unexamined.

Finally, it should be stated at the outset that a rigorous comparison may find that added complexity yields no significant improvement over a well-specified classical baseline. Such an outcome would be consistent with the econometric evidence and with the parallel findings in the tabular-data literature, and would constitute a legitimate and informative result rather than a failure. The value of the study lies in

the reliability of its conclusion, not in the direction of that conclusion. This principle carries into the methodology set out in the following chapter.

3 Methodology

This chapter sets out the methodological reasoning behind the study. It is organised in two parts. Section 3.1 analyses the principal threats to the validity of a volatility forecasting comparison, identifying the ways in which such a study can produce results that appear sound but are not. Section 3.2 then states the design choices adopted in response to each threat, with reference to the literature reviewed in Chapter 2. The implementation detail of how these choices were realised in code is deferred to Chapter 4; the concern here is with why the study is designed as it is.

The guiding principle throughout is that in an empirical forecasting comparison the credibility of the conclusion rests less on the sophistication of the models than on the integrity of the evaluation. A comparison conducted carelessly can favour whichever model the analyst expected to win, and do so invisibly. The design choices below are therefore directed almost entirely at removing avenues through which an unfair or unreliable result could arise.

3.1 Problem Analysis

Five threats to validity are identified. Each is capable, on its own, of undermining the study's central comparison.

3.1.1 Look-ahead bias and data leakage

The most serious threat in any time-series forecasting study is data leakage: the inadvertent use of information during model construction that would not have been available at the point of forecast. In a financial setting this is often termed look-ahead bias. Its effect is to produce performance figures that are optimistic and irreproducible in genuine out-of-sample use, because the model has been given access to the future. The most familiar form is the random partitioning of data into training and test sets, which is standard in cross-sectional machine learning but invalid for time series, since it permits a model to be trained on observations that occur chronologically after those on which it is tested (Bergmeir & Benitez, 2012).

Leakage also arises in less conspicuous forms, and it is these that most often escape notice. Normalising or scaling features using statistics computed over the entire sample allows information from the test period to influence the training data. Estimating model parameters on the full series before evaluating on a subset of it has the same effect. Selecting hyper-parameters using data that overlaps the test period leaks information through the tuning process rather than the training process.

Any one of these is sufficient to invalidate the comparison, and because each produce better apparent performance, none announces itself through poor results.

3.1.2 The unobservable target

Volatility is a latent quantity and is never directly observed, a point established in Section 2.2. The consequence for evaluation is that a forecast can only ever be compared against a proxy for the true volatility, and the quality of any accuracy assessment is bounded above by the quality of that proxy. Where the proxy is very noisy, as squared daily returns are, genuine differences in forecast accuracy between models may be indistinguishable from the noise of the proxy itself (Andersen & Bollerslev, 1998). The choice of both the target and the loss function used to evaluate against it is therefore not a matter of convenience but a determinant of whether the study can detect a real difference at all (Patton, 2011).

3.1.3 Baseline fairness

The study's conclusion depends entirely on the classical baseline being a fair representation of what classical models can achieve. A GARCH specification that is misconfigured, under-specified, or estimated incorrectly would constitute a weak benchmark, and demonstrating that a deep model outperforms a weak benchmark establishes nothing of interest. This threat is heightened in the present case because the author's background is in machine learning rather than econometrics. It is addressed directly in the design by validating the estimated baseline against results published in the literature, so that the benchmark's correctness can be verified independently rather than assumed.

3.1.4 Overfitting under limited data

Deep learning models contain a large number of parameters and are correspondingly prone to overfitting when trained on limited data. Daily financial data is limited in a way that is easy to underestimate: two decades of observations amount to only around five thousand data points, and within that span the number of genuinely distinct volatility regimes is small. A model may therefore appear to perform well by fitting features of the particular sample rather than the underlying process. A related concern is that the performance of a deep model can vary appreciably from one training run to another purely as a result of random initialisation, so that a single reported result may reflect a fortunate seed rather than a reliable capability.

3.1.5 Regime dependence

The relative performance of volatility models is unlikely to be constant across market conditions. Classical models may perform adequately in calm periods, while the behaviour of both classical and deep models during crises and regime transitions is

of particular interest and practical importance. A single performance figure aggregated over the whole sample risks concealing this variation, and could report an overall winner that is in fact inferior during precisely the turbulent periods when accurate volatility forecasts matter most.

3.2 Design Choices

The design of the study responds to each of the threats above. The choices are stated here as reasoned responses; their implementation is described in Chapter 4.

3.2.1 Walk-forward evaluation

To eliminate look-ahead bias, the study adopts walk-forward evaluation with an expanding origin rather than a random or single fixed split. At each step the models are estimated using only data available up to that point and are used to forecast the next observation, after which the origin advances and the process repeats. This reproduces the conditions of genuine forecasting, in which only the past is known (Bergmeir & Benitez, 2012). All data-dependent transformations, including feature scaling and model estimation, are confined to the data available at each origin, so that no information from the future can enter through preprocessing. The subtle leakage channels identified in Section 3.1.1 are thereby closed by construction rather than by inspection after the fact.

3.2.2 Realised volatility as the target

In view of the unobservable-target problem, the study uses realised volatility, constructed from high-frequency intraday returns, as the forecasting target rather than squared daily returns. Realised volatility aggregates many intraday observations into a substantially less noisy estimate of the day's true variance, and its use as an evaluation target is what allowed earlier work to demonstrate that standard volatility models forecast well after all (Andersen & Bollerslev, 1998). This choice raises the study's ability to detect genuine differences between models above what would be possible with a noisier proxy.

3.2.3 A fixed evaluation harness, specified before modelling

The evaluation procedure, comprising the target, the loss functions and the significance test, is specified and implemented before any model is built. This ordering is deliberate. If the evaluation were designed after the models, there would be scope, however unintentional, to select an evaluation that favoured a preferred model. Fixing the harness in advance removes that scope and is itself a commitment to a fair comparison. The primary loss function is QLIKE, chosen because it is robust to noise in the volatility proxy and because it penalises under-prediction of volatility more heavily than over-prediction, which is the appropriate asymmetry in a risk

context (Patton, 2011). The root mean squared error is reported alongside it for comparability with prior studies.

3.2.4 Statistical significance testing

A difference in average loss between two models does not by itself establish that one forecasts more accurately than the other, since the difference may reflect sampling variation. The study therefore applies the Diebold-Mariano test of equal predictive accuracy to every pairwise comparison of models (Diebold & Mariano, 1995). Reporting the significance of differences, rather than their magnitude alone, guards against drawing conclusions from margins that are within the range of noise, a shortcoming identified in much of the comparative literature in Section 2.7.

3.2.5 A validated classical baseline

To ensure baseline fairness, the classical models are not treated as a formality but are specified with care, including the asymmetric variants that the literature indicates are necessary for equity data, and the estimated GARCH(1,1) parameters are compared against values published for the same index in prior studies. Agreement provides independent evidence that the baseline has been constructed correctly, which is a precondition for the comparison to carry any weight. This step also responds directly to guidance from the project supervisor.

3.2.6 Multiple-seed reporting

To address the instability of deep models under limited data, each deep model is trained several times with different random initialisations, and its performance is reported as a distribution across seeds rather than as a single figure. Where the variation across seeds is comparable to or larger than the difference between a deep model and the classical baseline, any apparent advantage is treated as unreliable. This makes the stability of a result, and not merely its central value, part of the evidence.

3.2.7 Regime-disaggregated evaluation

Rather than reporting a single aggregate figure, the study evaluates performance separately across distinct market regimes: a calm period, the global financial crisis of 2008 to 2009, and the market disruption of early 2020. These two crises provide well-separated and clearly defined episodes of extreme volatility against which the models can be tested. This disaggregation exposes any dependence of model ranking on market conditions and allows the study to report not only which model forecasts best on average but whether that ordering holds when accurate forecasts are most needed. It should be noted that the realised volatility data used as the forecasting target extends to February 2022, a consequence of the discontinuation of

the source library; the sample is correspondingly bounded, and later periods such as the 2022 drawdown fall outside it.

3.2.8 Scope boundary

Finally, the study is deliberately bounded to the univariate forecasting of a single series, the S&P 500. Modelling the joint volatility of several assets would require multivariate techniques such as DCC-GARCH and multivariate deep architectures, which constitute a substantially larger undertaking. This boundary was agreed with the project supervisor as appropriate for the scope of the dissertation, and multivariate extension is recorded as future work in Chapter 6. Restricting the scope in this way concentrates effort on the rigour of the comparison rather than its breadth, which is where the contribution of the study lies.

3.3 Model Specifications

This section defines the models compared in the study. It introduces each briefly and states the key equation that characterises it, so that the results in Chapter 5 can be understood in terms of what each model actually computes. The models are presented in the order of the ladder, from the simplest baseline to the deep and hybrid architectures. Throughout, r_t denotes the log return on day t , and the conditional variance of that return, the quantity being forecast, is written as σ_t^2 . Fuller treatments of the classical volatility models summarised here can be found in standard references on financial time series (Tsay, 2010).

3.3.1 Exponentially weighted moving average (EWMA)

The EWMA model, in the RiskMetrics form, estimates the conditional variance as a geometrically weighted average of past squared returns, so that recent observations carry more weight than distant ones. It has a single parameter, the decay factor λ , conventionally fixed at 0.94 for daily data. It serves as the naive benchmark of the ladder.

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1-\lambda) r_{t-1}^2$$

3.3.2 GARCH(1,1)

The generalised autoregressive conditional heteroskedasticity model of Bollerslev (1986) makes the conditional variance an explicit function of both the previous period's squared shock and the previous period's variance. The first term, ω , is a constant; the second, the ARCH term, captures the reaction of volatility to recent market shocks; the third, the GARCH term, captures the persistence of volatility over time. The sum α plus β measures how slowly the effect of a shock decays, and for daily equity data is typically close to one.

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

Here ε_{t-1} is the return innovation (the unexpected component of the previous return), and the parameters are constrained so that ω is positive and α and β are non-negative, which guarantees a positive variance.

3.3.3 Asymmetric extensions: EGARCH and GJR-GARCH

The basic GARCH model treats positive and negative shocks identically and so cannot capture the leverage effect, the empirical tendency of negative returns to raise volatility more than positive returns of equal size. Two extensions address this. The GJR-GARCH model of Glosten, Jagannathan and Runkle (1993) adds a term that activates only when the previous shock was negative:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma I_{t-1} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$$

where the indicator I_{t-1} equals one when ε_{t-1} is negative and zero otherwise, so that gamma measures the additional impact of negative shocks. A positive gamma indicates the expected leverage effect. The EGARCH model of Nelson (1991) instead models the logarithm of the variance, which guarantees positivity without parameter constraints and allows the sign and magnitude of a shock to affect volatility separately:

$$\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \alpha(|z_{t-1}| - E|z_{t-1}|) + \gamma z_{t-1}$$

where z_{t-1} is the standardised innovation. In this parameterisation a negative gamma corresponds to the leverage effect. These two models are included because, as the literature indicates for equity data, asymmetry is essential rather than optional.

3.3.4 Heterogeneous autoregressive model (HAR-RV)

The heterogeneous autoregressive model of Corsi (2009) takes a different approach: rather than modelling conditional variance from returns, it models realised volatility directly as a linear regression on its own averages over three horizons, daily, weekly and monthly. The rationale is that market participants operate over different time scales, and the three components approximate their combined behaviour. Writing RV for realised volatility, the one-day-ahead forecast is:

$$RV_t = \beta_0 + \beta_d RV^{(d)}_{t-1} + \beta_w RV^{(w)}_{t-1} + \beta_m RV^{(m)}_{t-1} + \varepsilon_t$$

where the daily, weekly and monthly terms are averages of past realised volatility over the previous one, five and twenty-two trading days respectively. Despite its simplicity, the model is a strong and widely used benchmark for realised volatility forecasting, and in this study it is the most accurate of the classical models.

3.3.5 Long short-term memory network (LSTM)

The deep model used in this study is a long short-term memory network, a type of recurrent neural network introduced by Hochreiter and Schmidhuber (1997). A

recurrent neural network processes a sequence one step at a time, maintaining an internal state that summarises what it has seen so far, which in principle allows it to model temporal dependence of the kind seen in volatility. Simple recurrent networks are difficult to train over long sequences because the gradients used to fit them tend to vanish, so information from the distant past is lost. The long short-term memory network solves this with a gated memory cell: a set of learned gates control what information is added to the cell, retained in it, and read out from it, allowing the network to preserve relevant information across many time steps.

At each step the network updates a cell state c_t and a hidden state h_t through three gates, the forget gate f_t , the input gate i_t and the output gate o_t , each a function of the current input and the previous hidden state:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

where sigma is the logistic function. The cell state is updated by forgetting part of its previous value and adding new candidate information, and the hidden state is the gated output of the cell:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

$$h_t = o_t \odot \tanh(c_t)$$

where the symbol between terms denotes element-wise multiplication, and the matrices W and U and vectors b are the parameters learned during training. In this study the input x_t at each step comprises the recent history of realised volatility together with the exogenous VIX feature, and the network is trained to forecast the next day's volatility. Its relevance is that it can, in principle, capture nonlinear temporal patterns and incorporate exogenous information that the GARCH family cannot use, which is precisely the capability the study sets out to test.

3.3.6 GARCH-LSTM hybrid

The hybrid model combines the two approaches. It is the same long short-term memory network with one additional input feature: the one-step-ahead conditional variance forecast produced by the GARCH(1,1) model. The intention is that the econometric model supplies the structured volatility estimate it captures well, while the network models any remaining nonlinearity and incorporates the exogenous information. The construction of this feature is subject to strict controls against look-ahead bias, described in Section 4.8.

3.3.7 Evaluation metric: the QLIKE loss

Forecasts are scored with the QLIKE loss, which is robust to noise in the realised volatility proxy and penalises under-prediction of volatility more heavily than over-prediction. For a variance forecast h and a realised-variance proxy σ^2 it is defined as:

$$QLIKE = \sigma^2/h - \ln(\sigma^2/h) - 1$$

The loss attains its minimum of zero when the forecast equals the proxy and increases as the forecast departs from it, with the logarithmic term producing the asymmetry that makes it appropriate in a risk context.

4 Implementation

This chapter documents how the study was implemented. It describes the data sources and their provenance, the construction and validation of the forecasting target, the evaluation harness and the guarantees it provides, the implementation of each model in the ladder, and the software environment. The level of detail is intended to be sufficient for the study to be reproduced independently. All code, tests and result files are held in a version-controlled repository, and every reported figure is regenerable from the raw data by running the evaluation scripts.

4.1 Data Sources and Provenance

Three data sources were used. Daily price data for the SPDR S&P 500 exchange-traded fund, ticker SPY, and daily closing values of the CBOE Volatility Index, ticker VIX, were obtained through the yfinance interface to Yahoo Finance. The realised volatility series was obtained from the Oxford-Man Institute Realised Library (Heber et al., 2009).

The Oxford-Man library requires comment. It was discontinued in 2022 and its original distribution is no longer available, so the data was obtained from an archived copy of the complete library file, retrieved through the Internet Archive. The archived file was stored unmodified as the immutable raw artefact, and all filtering and cleaning performed in code. This arrangement has an incidental benefit for reproducibility: because the archived dataset is fixed and cannot be revised, the results reported here are exactly reproducible by any reader who obtains the same archived file. It also bounds the study, since the library ends on 25 February 2022. The source URL and access date for each dataset are recorded in the repository.

All three raw files were treated as immutable. No raw file was edited manually at any point; every transformation is performed in code from the raw artefacts, so that the processed dataset can be regenerated in full from source.

4.2 Target Variable Construction

The forecasting target is realised variance for the S&P 500, taken from the Oxford-Man library. The library file contains many indices in a single stacked table, identified by a symbol column, and provides several realised-variance estimators for each. The series used here is the symbol for the S&P 500 index, with the five-minute realised variance estimator. Both the symbol and the estimator are declared as named constants in the target-construction script and reported in its output at run time, so that the selection is auditable rather than implicit.

Daily logarithmic returns were computed from SPY closing prices. The realised variance series, the return series and the VIX series were then aligned on their common trading days by inner join, yielding a single processed dataset of 4,877 observations spanning 4 January 2000 to 25 February 2022. The realised variance series was converted to realised volatility, and to an annualised form, for reporting and visualisation; all model scoring is performed in variance space.

The constructed target was validated before use. The script asserts that dates are monotonic, that all realised volatility values are strictly positive, and that no daily return exceeds a plausible magnitude. It then verifies that the series exhibits the expected volatility spikes within known crisis windows, failing loudly if it does not. Both crisis regimes were confirmed to be well populated, with 138 trading days in the 2008 window and 46 in the 2020 window, and to show peak realised volatility of 12.8 and 9.3 times the sample median respectively.

4.3 Feature Set

Three per-timestep input features were used across the deep models. The first is the logarithm of realised variance, which supplies the autoregressive signal, being the same information the heterogeneous autoregressive model exploits. The second is the VIX, included as an exogenous feature. The rationale for its inclusion is that implied volatility is forward-looking and therefore carries information about anticipated market stress which a backward-looking realised volatility series cannot contain, and which the GARCH framework has no native mechanism to use. The third feature, used only by the hybrid model, is the logarithm of a GARCH-derived conditional variance forecast, described in Section 4.7.

The feature set is declared at model construction rather than embedded in the model code, which permits a feature to be removed without any change to the model implementation. This is what makes the ablation studies of Chapter 5 a matter of configuration rather than of rewriting, and ensures that an ablated model differs from its comparator in the feature set alone.

4.4 Exploratory Analysis

Exploratory analysis was conducted on the processed dataset before modelling. Plots of daily returns and of squared returns, shown in Figure 4.1 and Figure 4.2, display the volatility clustering described in Section 2.1. The most directly informative diagnostic, shown in Figure 4.3, is the comparison of the autocorrelation function of returns with that of squared returns: the former is close to zero at all lags while the latter is positive and decays slowly, which is the empirical basis for the entire GARCH family and for treating volatility rather than direction as the forecastable quantity. A quantile-quantile plot against the normal distribution, shown in Figure 4.4, confirms the heavy tails discussed in Section 2.1, and a plot of the VIX against realised volatility, shown in Figure 4.5, illustrates the relationship motivating its use as an exogenous feature.

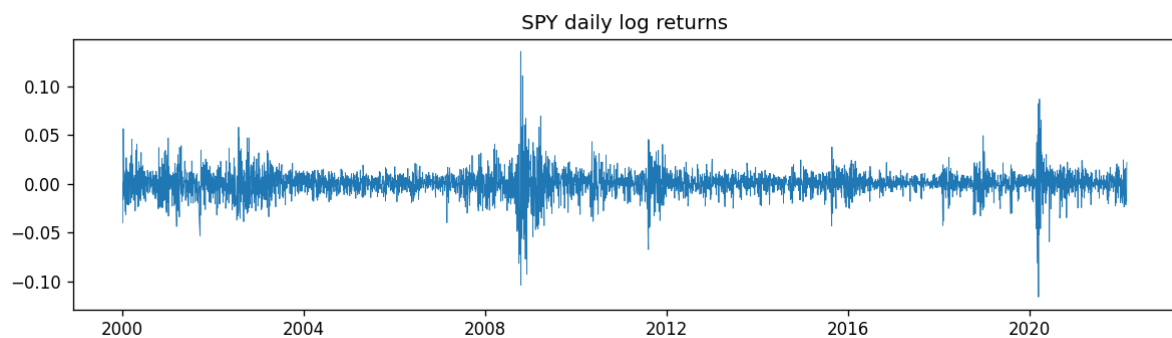


Figure 4.1. SPY daily log returns, 2000 to 2022. The clustering of large movements in 2008 and 2020 is visible directly.

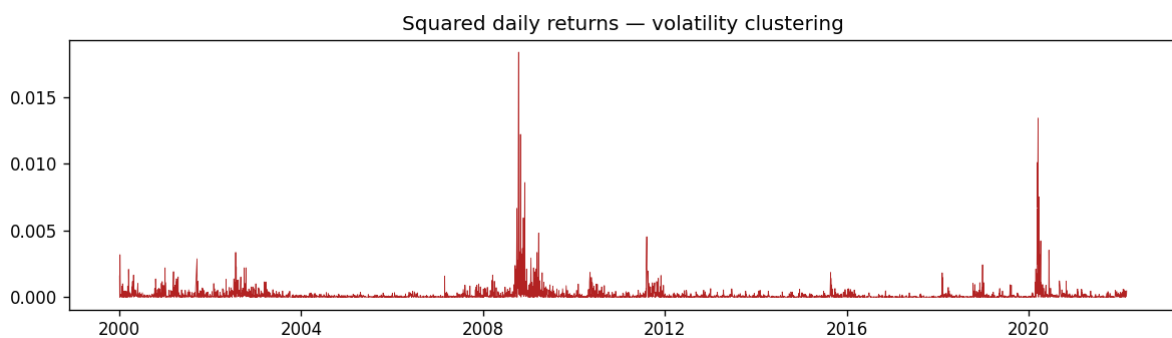


Figure 4.2. Squared daily returns, 2000 to 2022, illustrating volatility clustering: large squared returns are grouped in time rather than dispersed.

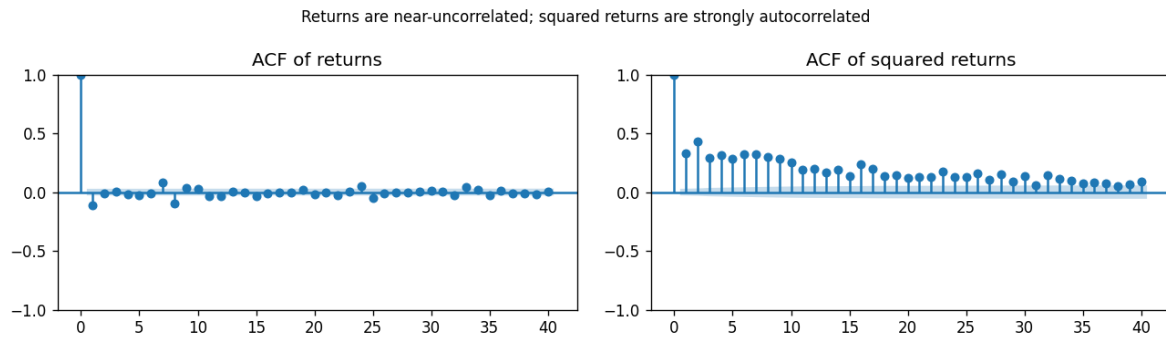


Figure 4.3. Autocorrelation of returns (left) against autocorrelation of squared returns (right). Returns are near-uncorrelated at all lags while squared returns are strongly and persistently autocorrelated. This contrast is the empirical basis for the GARCH family and for treating volatility, rather than direction, as the forecastable quantity.

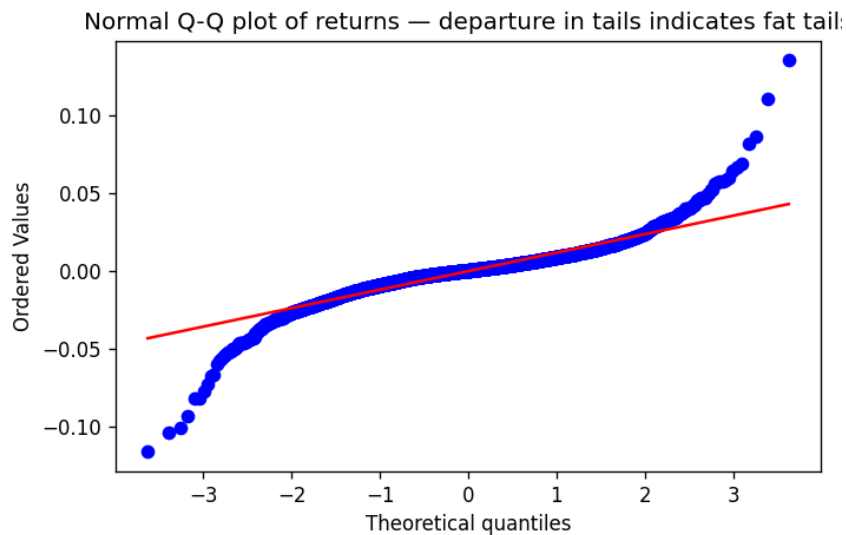


Figure 4.4. Normal quantile-quantile plot of daily returns. The departure from the reference line in both tails indicates the heavy tails characteristic of financial returns.

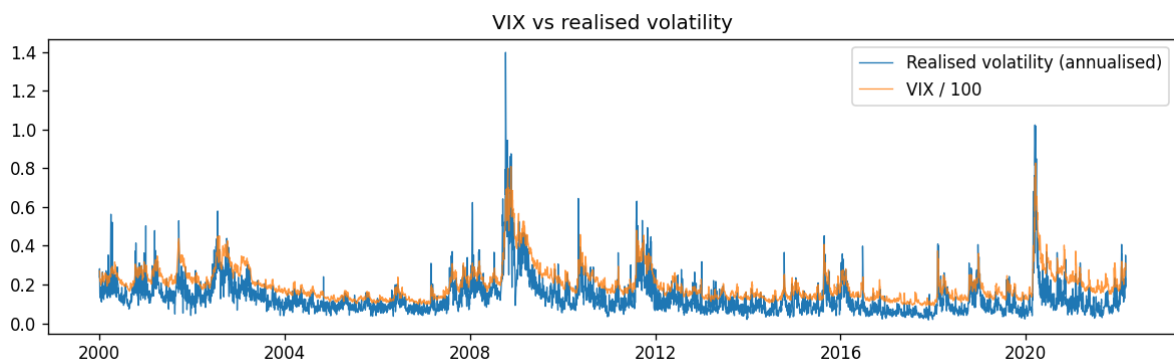


Figure 4.5. The VIX plotted against annualised realised volatility, 2000 to 2022. The two series move together, motivating the inclusion of the VIX as an exogenous feature.

4.5 The Evaluation Harness

The evaluation harness was implemented before any forecasting model, for the reasons given in Section 3.2.3. It defines a single interface that every model implements, consisting of a method to estimate parameters from a training window and a method to produce a one-step-ahead variance forecast. All models therefore pass through identical evaluation code.

Forecasting proceeds by expanding-origin walk-forward. At each origin the harness slices the dataset up to and including that origin and passes only that slice to the model; the model produces a variance forecast for the following day; and the harness then reads the realised value for that day solely in order to score the forecast. The first forecast is made for 2 January 2004, following an initial training window of 872 observations, and the procedure yields 4,005 forecasts through to the end of the sample. The same 4,005 origins are used for every model, including both arms of each ablation, so that all comparisons are made on identical data.

Prevention of look-ahead bias is structural rather than procedural. Because the harness performs all slicing and passes each model only the observations available at the origin, a model cannot access a future observation for the reason that it is never given one. The harness additionally asserts at every iteration that the last date visible to the model precedes the date being forecast, and validates that each returned forecast is a strictly positive finite variance, raising an error naming the model and date if not. All scaling and normalisation is fitted by each model on its training window, within the slice supplied; no statistic is ever computed over the full series.

The QLIKE and root mean squared error losses, the regime assignment, and the Diebold-Mariano test were implemented within the harness. The Diebold-Mariano test is computed on the per-observation QLIKE loss differentials, using a Newey-West long-run variance estimator with the Harvey, Leybourne and Newbold small-sample correction, and is reported both overall and within each regime.

4.6 Classical Model Implementation

The EWMA baseline follows the RiskMetrics specification, with the decay parameter fixed at 0.94. It was deliberately not tuned: as the naive benchmark of the ladder, optimising it would compromise the interpretation of the comparison.

The GARCH(1,1), EGARCH and GJR-GARCH models were estimated by maximum likelihood using the arch library, re-fitted at every one of the 4,005 forecast origins in accordance with the strict protocol. Every fit converged and every fit satisfied the stationarity condition. The estimated parameters were retained at each origin and are reported in Chapter 5 as the baseline validation.

The heterogeneous autoregressive model was implemented as an ordinary least squares regression of realised volatility on its own daily, weekly and monthly averages, following Corsi (2009). Because this model predicts realised volatility from lagged realised volatility, the alignment of its predictors was given particular attention: the most recent value entering any regression is from the day preceding the target, and this boundary is asserted in the test suite.

4.7 Deep Model Implementation

The full configuration of the deep models is summarised in Table 4.1. The network is a single-layer long short-term memory model with 32 hidden units, followed by a linear output layer producing a scalar. It reads sequences of 22 trading days, matching the monthly horizon of the heterogeneous autoregressive model so that the two receive comparable history and any difference in performance is attributable to how that history is used rather than to how much of it is seen. The architecture was deliberately kept small, given the limited number of observations available relative to the parameter count and the consequent risk of overfitting.

The network predicts standardised logarithmic realised variance. Working in logarithms both stabilises the target distribution and guarantees, through the exponential inverse transform, that the recovered variance forecast is strictly positive, so no flooring of forecasts is required. Training minimises mean squared error on the standardised logarithmic target using the Adam optimiser (Kingma & Ba, 2015) with a learning rate of 0.001 and a batch size of 64, for a maximum of 100 epochs with early stopping after 10 epochs without improvement. The validation set used for early stopping is the most recent 15 per cent of the training window by date, which lies entirely before the forecast origin. All hyper-parameters were fixed in advance and were not tuned against the test period.

Setting	Value
Architecture	1 LSTM layer, 32 hidden units, linear output
Lookback window	22 trading days
Target	standardised log realised variance
Loss	mean squared error
Optimiser	Adam, learning rate 0.001
Batch size	64
Maximum epochs	100 (early stopping, patience 10)
Validation split	most recent 15 per cent of training window
Random seeds	5 (0 to 4)
Re-estimation	every 21 origins (monthly)

Table 4.1. Deep model configuration.

Standardisation statistics are computed on the training window at each re-estimation and applied to both the training sequences and the forecast input. One consequence of re-estimating less frequently than every origin deserves comment. Between re-estimations, the input window for a forecast may include observations that postdate

the fitting of the standardiser. This does not constitute leakage: the standardising constants derive only from data preceding the last re-estimation, and applying constants derived from training data to subsequent observations is the standard and correct procedure. The network weights likewise derive only from data preceding the last re-estimation and cannot encode later observations.

Because the network is stochastic, each configuration was trained under five random initialisations, with the random number generators of both the deep learning framework and the numerical library seeded per fit. Results are reported as the mean and standard deviation across seeds, and the significance tests are computed on the seed-averaged forecasts. The classical models are deterministic and were run once.

4.8 The Hybrid Model and its GARCH Feature

The hybrid model is the same network with a third input feature, rather than a separate architecture. This is a deliberate implementation choice: because the hybrid differs from the network only in its feature set, the ablation that removes the GARCH feature isolates the contribution of that feature alone.

The construction of the GARCH feature is the most leakage-sensitive element of the study, and it was implemented accordingly. The feature for a given day is the one-step-ahead conditional variance forecast produced by a GARCH(1,1) model estimated on data up to the preceding day. It is generated by the same walk-forward code path used to evaluate the standalone GARCH model, with strict re-estimation at every origin, so that the leakage guarantees verified for that path apply identically here. It is emphatically not the in-sample fitted conditional variance of a model estimated over the full sample, which would incorporate future information and would render the hybrid results meaningless.

The feature requires a burn-in period of 250 trading days before the first stable estimate, so the series begins in February 2001. Because this precedes the first forecast origin of January 2004, the burn-in imposes no loss of scored observations, and both hybrid arms are evaluated on the same 4,005 origins as every other model. The feature is cached to avoid recomputation; this caching is licensed by an explicit test demonstrating that the column is fully determined by past data, rather than being assumed safe.

4.9 Verification and Testing

The implementation is covered by a suite of 62 automated tests, all of which pass. These include unit tests of the loss functions against hand-computed values, tests that the QLIKE loss is minimised when the forecast equals the proxy and that it penalises under-prediction more heavily than over-prediction, tests of the Diebold-

Mariano statistic under known conditions, and tests of regime assignment and walk-forward alignment.

Of greater importance to the credibility of the study are the adversarial leakage tests. Models were written for the express purpose of attempting to read the observation being forecast, both by position and by date, and both fail with errors demonstrating that the target row is absent from the data supplied to the model. For the deep models, a test confirms that no input window contains its own target day, and that corrupting all data from the target day onward leaves the forecast unchanged. For the hybrid, three tests establish the integrity of the GARCH feature: that a column constructed on a truncated prefix of the data reproduces the full-sample values exactly on the overlap, demonstrating that no future information enters; that the column differs materially from the full-sample fitted conditional variance, demonstrating that the forbidden quantity is not what is produced; and that corrupting data from a given day onward leaves all earlier feature values unchanged.

Further tests protect the integrity of the comparison itself, confirming that an ablated model differs from its comparator only in the number of features, and that changes made to support the hybrid leave the previously computed network results bit-identical.

4.10 Software Environment and Reproducibility

The study was implemented in Python 3.11 using the arch library (Sheppard et al., 2024) for the classical volatility models, PyTorch 2.3.1 for the deep models, and statsmodels 0.14.2, pandas 2.2.2, numpy 1.26.4, scipy 1.13.1 and scikit-learn 1.5.1 for supporting computation. Versions are pinned in the repository. All computation was performed on a central processing unit. Model runtimes are themselves recorded as a reproducible artefact: a timing script re-runs the classical models through the same harness and writes the measured wall-clock times to a results file, so that the figures reported in Chapter 5 are regenerable rather than transcribed from console output. Absolute times are hardware-dependent; the per-origin costs and the ratios between models are the portable quantities.

The repository is organised so that raw data, processed data, source code, tests and results are separated, with raw and processed data excluded from version control as regenerable artefacts while their provenance is documented. Each stage of the work was committed as it was completed, with the result of that stage recorded in the commit message, so that the development history is itself an auditable record of the study.

5 Presentation of Results

This chapter presents the empirical results of the study. It reports the forecast accuracy of each model in the ladder, the statistical significance of the differences between them, the variation of performance across market regimes, and the ablation results that decompose the sources of forecast skill. Interpretation of these findings, and their implications for the research question, is reserved for Chapter 6; the present chapter is concerned with reporting what was observed.

5.1 Evaluation Metrics

Two loss functions are reported throughout. The primary measure is the QLIKE loss, computed in variance space, which is robust to noise in the realised-volatility proxy and penalises under-prediction of volatility more heavily than over-prediction, an asymmetry appropriate to a risk context. The QLIKE loss for a variance forecast h against a realised-variance proxy σ^2 is given by σ^2 divided by h , minus the natural logarithm of σ^2 divided by h , minus one. It attains its minimum of zero when the forecast equals the proxy. The root mean squared error is reported alongside QLIKE, computed in volatility space for interpretability. For both losses a lower value indicates a more accurate forecast.

The statistical significance of the difference in accuracy between any two models is assessed using the Diebold-Mariano test of equal predictive accuracy, applied to the per-observation QLIKE loss differentials, with a Newey-West long-run variance estimator and the Harvey, Leybourne and Newbold small-sample correction. By the sign convention adopted, a positive test statistic indicates that the first-named model has the higher loss and is therefore the less accurate of the pair.

5.2 Overall Forecast Accuracy

Table 5.1 reports the overall forecast accuracy of all seven model configurations across the full out-of-sample period, comprising 4,005 daily forecasts from January 2004 to February 2022. The models are ordered as they appear in the ladder. Results for the stochastic deep models are reported as the mean across five random seeds, with the standard deviation given in brackets.

Model	QLIKE	RMSE (vol)
EWMA	0.35429	0.00519
GARCH(1,1)	0.35502	0.00475
GJR-GARCH	0.33288	0.00461
EGARCH	0.32401	0.00397
HAR-RV	0.21601	0.00330
LSTM	0.2137 (0.0013)	—
GARCH-LSTM hybrid	see 5.6	—

Table 5.1. Overall forecast accuracy of all models (4,005 forecasts, 2004-2022). Lower is better. Deep models reported as mean across five seeds.

Among the classical models, the naive EWMA baseline and the symmetric GARCH(1,1) are almost indistinguishable, at 0.354 and 0.355 respectively. The two asymmetric specifications improve on both, with EGARCH (0.324) ahead of GJR-GARCH (0.333). The heterogeneous autoregressive model achieves the lowest classical loss by a substantial margin at 0.216, a reduction of approximately one third relative to the best GARCH-family model.

The long short-term memory network achieves an overall QLIKE of 0.2137, marginally below the heterogeneous autoregressive model's 0.2160. As Section 5.3 reports, this nominal difference is not statistically significant. The hybrid model likewise does not improve on the heterogeneous autoregressive benchmark overall. No deep configuration significantly outperforms the strongest classical model. The forecasts of the leading models across the full sample are shown in Figure 5.2.

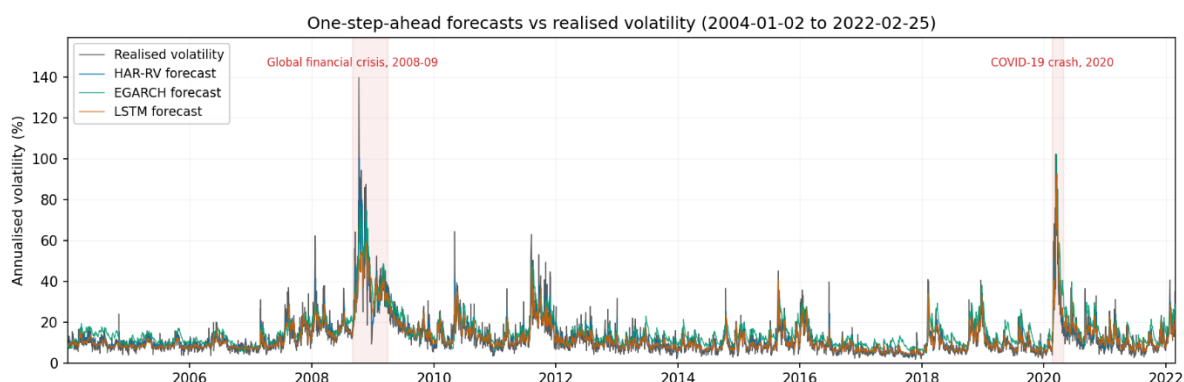


Figure 5.2. One-step-ahead forecasts of the leading models against realised volatility over the full out-of-sample period, 2004 to 2022, with the two crisis windows shaded. The models track realised volatility closely throughout, with visible divergence concentrated in the crisis periods.

5.3 Statistical Significance

Table 5.2 reports the Diebold-Mariano statistics and *p*-values for the comparisons of principal interest. A negative statistic favours the first-named model.

Comparison	DM statistic	p-value	Interpretation
EGARCH vs GARCH(1,1)	—	<0.0001	EGARCH significantly better
GJR vs GARCH(1,1)	—	<0.0001	GJR significantly better
EGARCH vs GJR	—	0.002	EGARCH significantly better
HAR vs EGARCH	+14.38	<0.0001	HAR significantly better
HAR vs LSTM	+1.147	0.25	Not significant (tied)
HAR vs hybrid	—	0.71	Not significant (tied)

Table 5.2. Selected pairwise Diebold-Mariano tests, overall period. Full pairwise matrix in Appendix [X].

Three findings emerge. First, both asymmetric models improve significantly on the symmetric GARCH(1,1) and on the EWMA baseline, and EGARCH improves significantly on GJR-GARCH; the advantage of asymmetric specifications is therefore not attributable to sampling variation. Second, the heterogeneous autoregressive model improves significantly on EGARCH over the full period. Third, and centrally for the research question, neither the long short-term memory network ($p = 0.25$) nor the hybrid ($p = 0.71$) differs significantly from the heterogeneous autoregressive model. The deep models reach the classical benchmark but do not surpass it.

5.4 Results by Market Regime

Table 5.3 disaggregates the QLIKE loss across the three market regimes: the calm periods constituting the majority of the sample, the global financial crisis of 2008 to 2009, and the market disruption of early 2020.

Model	Calm (n=3,821)	GFC 2008 (n=138)	COVID 2020 (n=46)
EWMA	0.34814	0.35602	0.85992
GARCH(1,1)	0.35494	0.28572	0.56977
GJR-GARCH	0.33367	0.25074	0.51377
EGARCH	0.32860	0.19336	0.33448
HAR-RV	0.21333	0.20027	0.48650
LSTM	0.2097 (0.0011)	0.2720 (0.0160)	0.3666 (0.0130)

Table 5.3. QLIKE loss by market regime. Deep models reported as mean (standard deviation) across five seeds.

The disaggregation reveals structure that the overall figures conceal. The heterogeneous autoregressive model owes its classical dominance almost entirely to the calm regime, where it achieves 0.213 against EGARCH's 0.329, since it models realised volatility directly and tracks it closely under normal conditions. In the crisis regimes the ordering changes: during the disruption of early 2020 EGARCH records the lowest classical loss (0.334) ahead of the heterogeneous autoregressive model (0.487).

The long short-term memory network shows an uneven crisis profile. It performs comparably to the heterogeneous autoregressive model in calm markets, is significantly worse during the 2008 crisis (Diebold-Mariano $p = 0.019$), and is marginally better during the 2020 disruption ($p = 0.087$). Its aggregate parity with the classical benchmark therefore conceals divergent behaviour across regimes. The behaviour of the models during the two crisis periods is shown in Figure 5.3 for the 2008 global financial crisis and Figure 5.4 for the COVID-19 disruption of 2020.

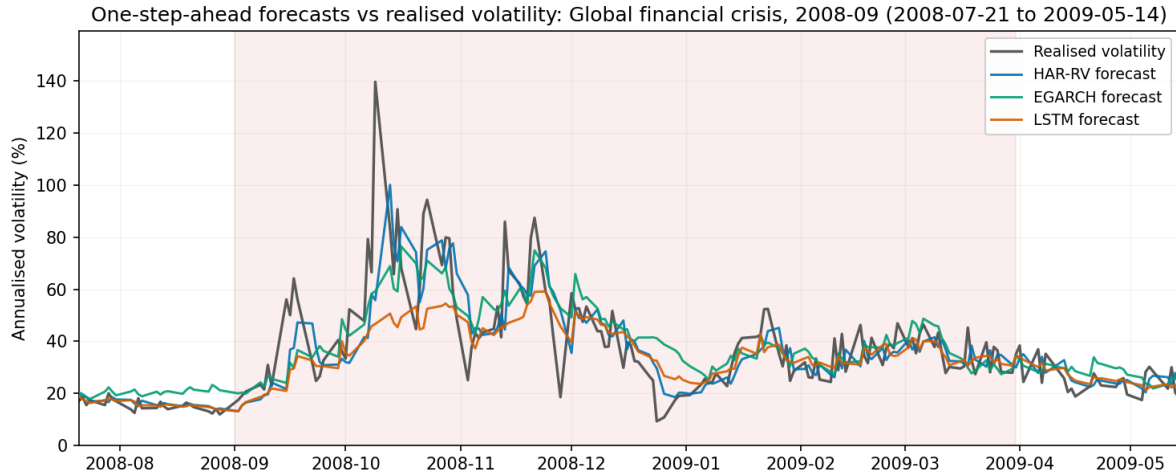


Figure 5.3. One-step-ahead forecasts against realised volatility during the global financial crisis of 2008 to 2009 (shaded). The heterogeneous autoregressive and asymmetric GARCH models track the crisis comparably, while the network responds less sharply to the largest spikes.

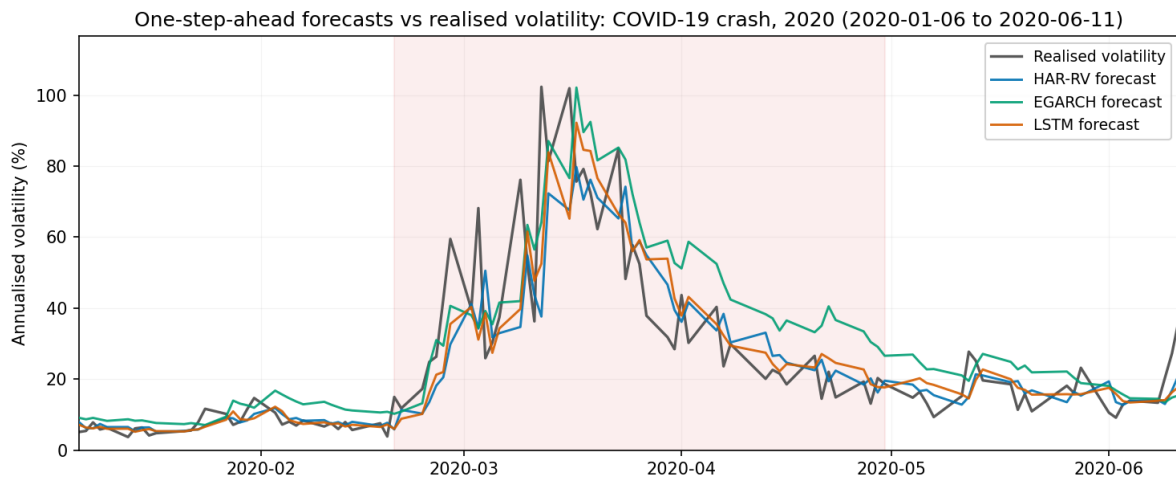


Figure 5.4. One-step-ahead forecasts against realised volatility during the COVID-19 disruption of early 2020 (shaded). EGARCH remains elevated through the recovery while realised volatility decays, illustrating the tendency of the asymmetric model to overshoot in the aftermath of an abrupt shock.

5.5 Stability Across Seeds

Because the deep models are stochastic, each configuration was trained under five random initialisations and its performance reported as a distribution rather than a single value. The long short-term memory network proved stable: its overall QLIKE across seeds was 0.2137 with a standard deviation of 0.0013, a coefficient of variation of approximately 0.6 per cent. This variation is an order of magnitude smaller than the differences between models reported above, so the results are not attributable to favourable initialisation. Seed variation was larger within the crisis regimes, where fewer observations are available, but remained small relative to the between-model differences. The classical models are deterministic and require no equivalent reporting.

5.6 Ablation Results

Two ablations were conducted to decompose the sources of forecast skill in the deep models. In each case the ablated arm was trained on an identical set of observations to its comparator, so that the contrast isolates the removed feature rather than any difference in training sample.

The first ablation removes the VIX feature from the long short-term memory network, leaving it with realised-volatility history alone and therefore the same information available to the heterogeneous autoregressive model. Table 5.4 reports the effect.

Regime	Full LSTM	LSTM without VIX	DM	p-value
Overall	0.2137	0.2383	-6.92	<0.0001
Calm	0.2097	0.2284	-5.54	<0.0001
GFC 2008	0.2720	0.3945	-3.84	0.0002
COVID 2020	0.3666	0.5924	-2.67	0.011

Table 5.4. Effect of removing the VIX feature from the LSTM. QLIKE, mean across five seeds.

Removing the VIX feature degrades the network significantly in every regime, and most severely in the crises, where the QLIKE loss rises by approximately 45 per cent during the 2008 crisis and 62 per cent during the 2020 disruption. Deprived of the exogenous feature, the network records an overall QLIKE of 0.2383 against the heterogeneous autoregressive model's 0.2160, and the classical model is significantly the more accurate of the two (Diebold-Mariano statistic -6.72, $p < 0.0001$).

The second ablation removes the GARCH-derived feature from the hybrid, isolating the contribution of econometric structure. Table 5.5 reports the effect. Overall the effect is null: the Diebold-Mariano statistic is +0.80 with a p-value of 0.42, indicating no significant difference. The regime disaggregation again reveals more than the aggregate. The GARCH feature significantly improves accuracy during the 2008 crisis (Diebold-Mariano statistic -3.40, $p = 0.0009$), recovering the weakness the network displayed in that regime, while marginally degrading accuracy in calm markets. The net effect across the full sample is therefore approximately zero.

Regime	Hybrid	Hybrid without GARCH feature	DM	p-value
Overall	0.2148	0.2141	+0.80	0.42
Calm	0.2115	0.2097	+2.09	0.036
GFC 2008	0.2657	0.2918	-3.40	0.0009
COVID 2020	0.3368	0.3482	-0.61	0.55

Table 5.5. Effect of removing the GARCH-derived feature from the hybrid model. QLIKE, mean across five seeds.

The QLIKE values in this table are computed on the seed-averaged forecast, so that they and the Diebold-Mariano statistics describe the same quantity; they differ marginally from the per-seed mean reported for the hybrid in Table 5.1, without affecting the conclusion.

5.7 Baseline Validation

Because the credibility of the comparison depends on the classical models being correctly specified, the estimated parameters of the GARCH-family models were compared against values published for the S&P 500. The GARCH(1,1) persistence, given by the sum of the ARCH and GARCH coefficients, averaged approximately 0.985 across the 4,005 fits, within the range of 0.98 to 0.99 commonly reported for daily equity index data. The asymmetry parameters of both leverage models carried the expected sign in every fit, indicating that negative return shocks raise subsequent volatility more than positive shocks of equal magnitude, and their magnitudes were consistent with published estimates. Every fit converged and every fit was stationary. These checks provide independent evidence that the classical baseline is correctly specified, which is a precondition for the comparison to carry weight.

One minor and expected feature was noted. The estimated news-impact coefficients run slightly above canonical published values, with correspondingly lower persistence coefficients, such that overall persistence remains as expected. This is attributable to the crisis-heavy character of the 2000 to 2022 sample and to the use of Normal rather than Student-t innovations. Neither is a specification concern given that persistence, sign and convergence are all as expected.

5.8 Computational Cost

The models differ by several orders of magnitude in computational cost. All classical models were re-estimated at every one of the 4,005 forecast origins under the strict protocol, and their full walk-forward evaluations completed in seconds: 2.06 seconds for the EWMA baseline and 4.53 seconds for the heterogeneous autoregressive model, rising to 51.34 seconds for GARCH(1,1), 64.36 seconds for GJR-GARCH and 103.07 seconds for EGARCH, the most computationally demanding classical specification. The deep models are of an entirely different order. The hybrid specification required 13,002 seconds, approximately 3.6 hours, for its five-seed evaluation, and its ablated arm 11,785 seconds; both were obtained under monthly rather than per-origin re-estimation, since strict re-estimation would have been infeasible within the resources available. Construction of the GARCH feature required a further 65 seconds. All timings were measured on the same machine and are reproducible from the repository.

5.9 Forecast Visualisation

Figure 5.5 plots the EWMA forecast against realised volatility over the full out-of-sample period, with the two crisis windows shaded. It illustrates the characteristic behaviour of the naive baseline, tracking realised volatility closely in calm periods but lagging the abrupt spikes of 2008 and 2020, where the forecast peak arrives late and at a lower level before decaying slowly. Equivalent plots for the remaining models are provided in Appendix [X].

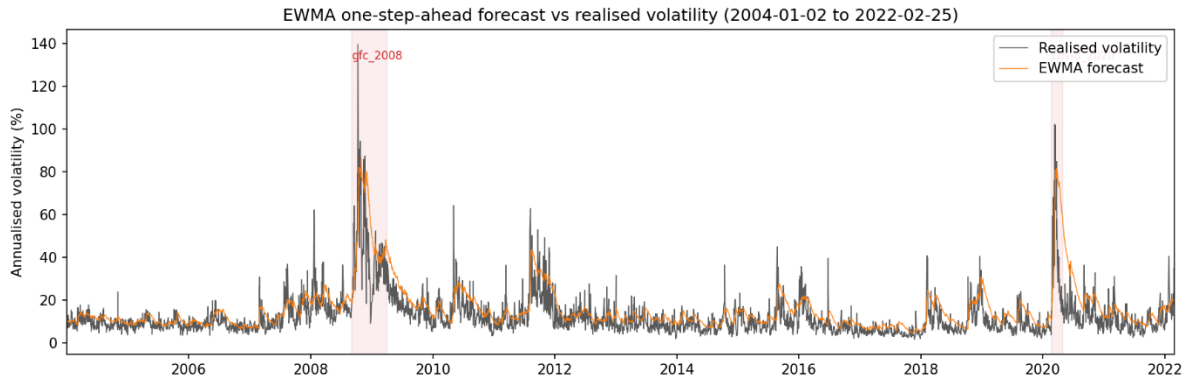


Figure 5.5. EWMA one-step-ahead forecast against realised volatility, 2004 to 2022, with crisis windows shaded. The baseline tracks realised volatility in calm periods but lags the abrupt spikes of 2008 and 2020.

6 Critical Review of Results

This chapter interprets the results reported in Chapter 5 and assesses what they establish in relation to the research question. It addresses in turn whether additional model complexity improved forecast accuracy, what the ablation studies reveal about the sources of that accuracy, how performance varied across market regimes, the practical question of computational cost, and the threats to the validity of the conclusions drawn.

6.1 Did Complexity Improve Forecast Accuracy?

The central question of this dissertation was whether deep learning models improve the accuracy of realised volatility forecasts relative to well-specified classical baselines, when evaluated under a protocol rigorous enough for the answer to be trusted. On the evidence presented, the answer is that they do not.

The long short-term memory network achieved an overall QLIKE of 0.2137 against the heterogeneous autoregressive model's 0.2160. Taken at face value this is a nominal improvement, and a study reporting point estimates alone might have presented it as such. The Diebold-Mariano test, however, returns a p-value of 0.25: the difference is well within the range attributable to sampling variation, and the two models are statistically indistinguishable. The hybrid specification, which adds an econometric structural feature to the network, likewise fails to separate from the

classical benchmark, with a p-value of 0.71. No configuration of the deep models significantly outperformed a simple linear regression on lagged realised volatility.

This finding should be interpreted carefully. It is not a claim that deep learning cannot forecast volatility, nor that a differently specified network might not perform better. The defensible claim is narrower and, for that reason, more secure: under a leakage-free walk-forward protocol, evaluated against a low-noise realised volatility proxy with a robust loss function and formal significance testing, a standard long short-term memory network does not improve on a correctly specified heterogeneous autoregressive model. Given that the difference in question carries a p-value of 0.25, and that the direction of the point estimate favours the deep model only marginally, it is improbable that modest additional tuning would alter this conclusion.

The result is consistent with the wider literature in two respects. It accords with the finding of Hansen and Lunde (2005) that well-specified classical volatility models are difficult to surpass, and it parallels the repeated observation in the tabular-data literature that carefully tuned classical methods match or exceed deep architectures on structured data (Shwartz-Ziv & Armon, 2022). What the present study adds is that this conclusion survives under an evaluation protocol specifically designed to remove the methodological weaknesses identified in Section 2.7.

6.2 Where Does Forecast Skill Come From?

The more informative result of this study is not that the deep models failed to win, but the decomposition of why they came as close as they did. Two ablations were conducted, and together they separate the contribution of model architecture from that of the information supplied to it.

The first ablation removed the VIX feature, leaving the network with realised-volatility history alone: precisely the information available to the heterogeneous autoregressive model. Under this condition the network's overall QLIKE deteriorated from 0.2137 to 0.2383, and the classical model became significantly the more accurate of the two, with a Diebold-Mariano statistic of -6.72 and a p-value below 0.0001. The implication is direct. On equal information, the flexible nonlinear architecture extracts less predictive content from realised-volatility history than a three-component linear regression does. The architecture is not merely failing to add value; it is performing worse than the simpler alternative it was intended to improve upon.

It follows that the network's parity with the classical benchmark in the unablated case is attributable entirely to the exogenous VIX feature. Removing that feature degrades performance significantly in every regime, and most severely in the crises, where the loss rises by approximately 45 per cent during the 2008 crisis and 62 per cent during

the 2020 disruption. This is economically intelligible: the VIX is a forward-looking measure of implied volatility, and it carries information about anticipated market stress that a backward-looking realised volatility series structurally cannot contain. The value lies in the information, not in the machinery that processes it.

The second ablation removed the GARCH-derived feature from the hybrid, isolating the contribution of econometric structure. Its aggregate effect was null, with a p-value of 0.42. The regime disaggregation, however, shows that this null conceals two offsetting effects: the feature significantly improved accuracy during the 2008 crisis, with a p-value of 0.0009, while marginally degrading it in calm markets. Econometric structure therefore redistributes forecast skill toward turbulent periods rather than raising it uniformly, and in this sample the two effects approximately cancel.

Taken together, the two ablations yield a decomposition that the comparative literature on deep volatility forecasting rarely provides. Architecture, evaluated on equal information, is a net liability. Exogenous information is the sole source of the deep models' competitiveness. Econometric structure alters the distribution of skill across market conditions without changing the aggregate. A study reporting only that a deep model matched a classical benchmark would have recorded none of this.

6.3 Regime Dependence

The disaggregation by market regime proved consistently more informative than the aggregate figures, and in several instances reversed the conclusion those figures would have supported.

Among the classical models, the heterogeneous autoregressive model's overall dominance derives almost entirely from calm markets, which constitute the large majority of the sample. In the crisis regimes its advantage disappears: during the 2020 disruption EGARCH recorded the lower loss, the reverse of the overall ordering. The explanation is structural. The heterogeneous autoregressive model forecasts from averages of past realised volatility, which respond slowly to an abrupt change, whereas the asymmetric GARCH models react directly to the large negative returns that characterise the onset of a crisis.

The deep models displayed an uneven crisis profile of their own. The network was significantly worse than the classical benchmark during the 2008 crisis yet marginally better during the 2020 disruption. The two episodes differ in character: the 2008 crisis developed over months, while the 2020 disruption was abrupt. That a single model should be relatively weak in one and relatively strong in the other indicates that its aggregate parity with the classical benchmark masks materially different behaviour under different conditions.

This is the empirical vindication of the design decision taken in Section 3.2.7. Had performance been reported only in aggregate, three findings of substance would have been lost: that the classical ordering reverses in crises, that the deep model's crisis behaviour is inconsistent, and that the GARCH structural feature is valuable in precisely the conditions where the aggregate suggests it is worthless.

6.4 Computational Cost

A finding of practical consequence concerns the disparity in computational cost. The heterogeneous autoregressive model completes its full walk-forward evaluation across 4,005 forecast origins in 4.53 seconds, with strict re-estimation at every origin. Even EGARCH, the most computationally demanding of the classical specifications, requires only 103 seconds under the same conditions. The deep models required several hours: the hybrid specification took 13,002 seconds, approximately 3.6 hours, across five seeds, and did so only under a relaxed monthly re-estimation schedule, since strict per-origin re-estimation would have demanded computation measured in days.

The deep models therefore consumed approximately 2,900 times the computation of the heterogeneous autoregressive model, and roughly 130 times that of the most expensive classical model, to achieve accuracy statistically indistinguishable from the former. The comparison is conservative, since the classical figures reflect strict re-estimation at every origin while the deep figures reflect a relaxed schedule. In a field increasingly disposed to adopt deep architectures by default, this asymmetry is worth stating plainly. Where a simple, interpretable and near-instantaneous model matches a complex and computationally expensive one, the burden of justification lies with the latter.

6.5 Threats to Validity

Several limitations qualify the conclusions above and are stated here in the interest of an accurate assessment of what the study establishes.

The most substantial concerns the scope of the architectural claim. The comparison evaluates one standard long short-term memory configuration, with hyper-parameters fixed a priori in order to avoid tuning against the test period. It is possible that a differently specified network, or a different deep architecture such as a temporal convolutional network or a transformer, would perform better. The claim defended here is accordingly that this architecture, under this protocol, does not improve on the classical benchmark, rather than that no deep architecture could. That said, the margin in the ablated comparison, with a Diebold-Mariano statistic of -6.72, is large enough that modest respecification would be unlikely to reverse it.

A second limitation concerns the re-estimation schedule. The classical models were re-estimated at every forecast origin, in accordance with the strict protocol, whereas the deep models were re-estimated monthly for reasons of tractability. This is a documented deviation. Its effect is to give the deep models slightly staler parameters than the classical models between re-estimations, which if anything works against them; the deviation is applied consistently across all deep configurations, so comparisons among them remain internally fair.

A third limitation concerns the sample. The realised volatility target extends to February 2022, a consequence of the discontinuation of the source library, and the study is confined to a single index. The findings are therefore established for the S&P 500 over 2004 to 2022 and should not be assumed to generalise to other asset classes, to markets with different microstructure, or to subsequent periods without further testing.

A fourth concerns the volatility proxy. Although realised volatility computed from five-minute returns is a substantially more accurate proxy than squared daily returns, it remains an estimate of a latent quantity rather than the quantity itself. The use of the QLIKE loss mitigates this, since it is robust to noise in the proxy (Patton, 2011), but no evaluation of volatility forecasts can escape the limitation entirely.

Set against these, several threats identified in Chapter 3 were successfully controlled. Look-ahead bias was eliminated by construction and verified by adversarial tests that attempt to access future observations and demonstrably fail. Baseline fairness was established by validating the estimated GARCH parameters against published estimates for the same index. Instability under random initialisation was addressed by reporting performance across five seeds, with a coefficient of variation of approximately 0.6 per cent, an order of magnitude smaller than the differences between models.

6.6 Potential Improvements

Several extensions would strengthen or broaden the present findings. Evaluating the same ladder across multiple assets and asset classes would establish whether the conclusions generalise beyond a single equity index. Extending the analysis to longer forecast horizons, such as five and twenty-two days, would test whether the relative standing of the models changes with horizon, as some literature suggests. Incorporating additional architectures, particularly attention-based models designed for multi-horizon forecasting with exogenous inputs, would widen the architectural claim. Finally, given that exogenous information proved to be the sole source of the deep models' competitiveness, an obvious extension is to supply the same implied-volatility information to the classical models, for instance by augmenting the

heterogeneous autoregressive specification with the VIX as an additional regressor. On the evidence of this study, that comparison would be the more interesting one.

7 Critical Review of Project Objectives

This chapter assesses the extent to which each of the objectives defined in Section 1.3 was met, including an account of the components that were modified or descoped during the project.

7.1 O1: Literature Review

The objective was to critically assess the methodological adequacy of existing comparisons between classical and deep learning approaches to volatility forecasting, and to establish a research gap. This objective was met. The review established the empirical properties that volatility models must accommodate, examined the consequences of volatility being a latent quantity, and traced the development of both model families. Its critical contribution was the identification in Section 2.7 of six recurring methodological weaknesses in the comparative literature, which together justified the study's central premise that the question remained open. The review also shaped concrete design decisions: the finding of Hansen and Lunde (2005) that asymmetry is essential for equity data determined the inclusion of EGARCH and GJR-GARCH, and the analysis of Patton (2011) determined the choice of loss function.

7.2 O2: Data and Target Construction

The objective was to acquire the necessary data and construct a validated realised volatility target. This objective was met, though not without a complication that required a change of approach. The Oxford-Man Institute Realised Library, the intended source of the realised volatility series, was discontinued in 2022, and the data was consequently obtained from an archived copy of the library with its provenance documented. This has the incidental advantage of fixing the dataset permanently, so that the results are exactly reproducible, but it bounds the sample at February 2022.

That constraint required a revision to the regime definitions. The originally planned analysis of the 2022 drawdown was not possible, since the target data covers only the first two months of that year. The regimes were accordingly redefined as calm periods, the global financial crisis of 2008 to 2009, and the market disruption of early 2020. This proved a superior design, providing two well-separated crisis episodes of contrasting character, both fully covered by the data. The target itself was validated against known volatility episodes before use.

7.3 O3: Evaluation Protocol

The objective was to construct a walk-forward evaluation harness free of look-ahead bias, employing the QLIKE loss and the Diebold-Mariano test, and to fix it before any model was built. This objective was met in full and represents the strongest element of the project. The harness was specified and implemented before the first model, which removed any possibility of an evaluation shaped, however unintentionally, to favour a preferred result.

Leakage prevention was implemented structurally rather than by convention: the harness performs all data slicing and passes each model only the observations available at the forecast origin, so that a model cannot access future data because it is never given it. This guarantee was verified adversarially. Models were written specifically to attempt to read the target observation, both by position and by date, and both failed with errors demonstrating that the future row was absent from the data supplied. The same verified code path was reused to construct the hybrid model's GARCH feature, ensuring that the most dangerous leakage risk in the project could not be reintroduced. The final test suite comprises 62 tests, all passing.

7.4 O4: Model Ladder

The objective was to implement a ladder of models of increasing complexity from an EWMA baseline through to a GARCH-LSTM hybrid. This objective was met in full: all six specified models were implemented and evaluated on identical forecast origins under the same harness.

The classical models were validated against the literature as required. The GARCH(1,1) persistence averaged approximately 0.985 across 4,005 fits, consistent with published estimates for daily equity index data; the asymmetry parameters of both leverage models carried the expected sign in every fit and lay within published ranges; and every fit converged. One deviation from the strict protocol was necessary. The deep models were re-estimated monthly rather than at every origin, because a single network fit requires seconds where a GARCH fit requires milliseconds, and strict re-estimation across five seeds would have required several days of computation. The deviation is documented, applied consistently, and works if anything against the deep models.

7.5 O5: Comparative Evaluation and Ablation

The objective was to evaluate all models on identical splits with significance testing, regime disaggregation, seed stability reporting and an ablation study. This objective was met in full. Every pairwise comparison was tested for statistical significance, results were disaggregated across three regimes, and the stochastic models were evaluated across five random seeds.

The ablation component proved more valuable than anticipated at the outset. It was originally conceived as a supporting analysis; in the event it produced the study's most substantive finding, namely that the deep model's competitiveness derives entirely from exogenous information rather than from its architecture. One methodological refinement was made during execution: the GARCH feature requires a burn-in period, which would have left the hybrid trained on fewer observations than its comparator and confounded the contrast with a difference in sample size. The ablation arm was therefore re-run on an identical set of observations, at the cost of additional computation, so that the comparison isolates the feature alone.

7.6 Overall Assessment

All five objectives were met. The principal changes from the original plan were the redefinition of the market regimes following the discovery that the target data ends in February 2022, and the relaxation of the re-estimation schedule for the deep models on grounds of computational tractability. Both are documented, and neither compromises the fairness of the central comparison. The project's strongest element is the evaluation protocol, which was specified in advance, verified adversarially, and applied identically to every model; its principal limitation is the scope of the architectural claim, which rests on a single standard network configuration and a single asset.

8 Conclusion

This dissertation asked whether deep learning models improve the accuracy of realised volatility forecasts relative to well-specified classical baselines, when the comparison is conducted under conditions rigorous enough for the answer to be trusted. The motivation was that the existing literature offers conflicting accounts, and that many of the comparisons on which those accounts rest exhibit methodological weaknesses which make their conclusions difficult to rely upon. The contribution sought was therefore methodological rather than architectural: not a new model, but a fair test.

The answer is that they do not. A long short-term memory network achieved an overall QLIKE loss of 0.2137 against the heterogeneous autoregressive model's 0.2160, a nominal improvement that the Diebold-Mariano test finds statistically insignificant at a p-value of 0.25. A hybrid specification incorporating an econometric structural feature likewise failed to separate from the same benchmark. No configuration of the deep models significantly outperformed a simple linear regression on lagged realised volatility, estimated in a fraction of a second.

The more informative result concerns the source of what accuracy the deep models did achieve. Two ablation studies decomposed their performance. Removing the

exogenous implied-volatility feature, leaving the network with exactly the information available to the classical benchmark, caused it to fall significantly behind that benchmark. On equal information, therefore, the flexible nonlinear architecture extracts less predictive content from realised-volatility history than a three-parameter linear regression does: the architecture is not merely failing to add value but is performing worse than the simpler alternative. It follows that the network's parity with the classical model in the unablated case derives entirely from the exogenous information supplied to it, and not from its architecture. The second ablation showed that econometric structure has no aggregate effect but redistributes forecast skill toward crisis periods, improving accuracy significantly during the 2008 crisis while marginally degrading it in calm markets.

These findings carry two implications. The first concerns the interpretation of reported improvements from deep learning in this domain. Where a deep model is supplied with information that its classical comparator does not receive, an observed improvement may reflect the information rather than the model, and the two possibilities have very different consequences for practice. Distinguishing between them requires ablation, which the comparative literature seldom performs. The second concerns cost. The classical benchmark completes a full walk-forward evaluation of more than four thousand forecasts in seconds; the deep models required several hours, and only under a relaxed re-estimation schedule, for accuracy that cannot be distinguished from it. Where a simple and interpretable model matches a complex and expensive one, the burden of justification lies with the latter.

The strength of these conclusions rests on the evaluation protocol rather than on the models. The forecasting target was realised volatility computed from high-frequency data rather than the noisy squared-return proxy that has historically distorted conclusions in this field. Evaluation was conducted by expanding-origin walk-forward, with look-ahead bias prevented by construction and verified by adversarial tests that attempt to access future observations and demonstrably fail. The classical baseline was validated against parameter estimates published for the same index, so that the comparison could not be dismissed as the defeat of a weak benchmark. Losses were measured with a proper scoring rule robust to proxy noise, differences were tested for statistical significance rather than asserted, stochastic models were reported across multiple random initialisations, and performance was disaggregated by market regime. It is this apparatus, specified before any model was built, that permits the null result to be stated with confidence rather than presented as an inconclusive outcome.

The limitations of the study should be restated. The architectural claim rests on one standard network configuration with hyper-parameters fixed in advance, and a

different architecture might perform differently, although the margin observed in the ablated comparison makes a reversal unlikely under modest respecification. The deep models were re-estimated monthly rather than at every origin for reasons of tractability, a documented deviation applied consistently and one which works if anything against them. The study covers a single equity index over the period 2004 to 2022, bounded at that date by the discontinuation of the realised volatility library, and the findings should not be assumed to generalise beyond it without further testing.

Several extensions follow naturally. Replicating the ladder across multiple assets and asset classes would establish the generality of the conclusions. Extending the analysis to longer forecast horizons would test whether the relative standing of the models is horizon-dependent. Incorporating attention-based architectures designed for multi-horizon forecasting with exogenous inputs would broaden the architectural claim. The most direct extension, however, follows from the central finding itself: since exogenous information proved to be the sole source of the deep models' competitiveness, the natural next comparison is one in which the same implied-volatility information is supplied to the classical models as well, for instance by augmenting the heterogeneous autoregressive specification with the VIX as an additional regressor. On the evidence presented here, that is the more interesting question.

The wider lesson of this work is not about volatility, and not about deep learning. It is that in applied machine learning the trustworthiness of a measurement matters more than the sophistication of the thing being measured. Every substantive finding reported here, from the insignificance of the deep models' apparent advantage to the discovery that their competitiveness came from information rather than architecture, was visible only because the evaluation was built to reveal it. Under a weaker protocol, each would have been reported as a modest success for deep learning, and each would have been wrong.

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Appendix A: Ethical Approval

Secondary Data Usage in Research Projects

If you are ONLY using secondary data for your research project, your research application may not need to be submitted through the ethics review process (Ethics RM).

Please read below to see if your study fits the criteria of research involving secondary data.

Usage

- Secondary data is data not collected specifically for the current study.
- It may be data from a public or subscription database, another organisation or data collected for a previous research project for which consent was given to use for future studies.

Please note that this does not include data publicly available on the internet (except those in public or subscription database or repositories) and data available on public social media feeds. Social media or other on-line research involving human participants is likely to require ethics approval and if so, this must be secured for a project before the collection of any data.

If your research involves higher sensitivity or risk topics or you are unsure in any way that your data is covered by the checklist, please discuss with your supervisor, and together you can email ethics ethics@surrey.ac.uk to ensure it meets the criteria.

Before using Secondary Data, you must:

- Discuss the data and project with your supervisor
 - See above about any higher sensitive or risk projects, or where you are unsure.
- Fill out the Secondary Data checklist on page 2 to see if the data can be used without going through the ethical review process on Ethics RM
 - If your data fulfils the criteria: ask supervisor to sign document as criteria met meaning no ethical review needed.
 - If the criteria are not met: ethical review required through ethical review process on Ethics RM (if unsure, consult with supervisor, who will contact ethics).
- If required for your dissertation (consult guidelines & supervisor), then add pdfs into your dissertation appendix of this signed secondary data checklist.

Staff: please follow the checklist and keep a copy of the checklist for audit purposes.

Secondary Data Checklist

Project Name: Beyond GARCH? A Rigorous Comparison of Classical and Deep Learning Models for Financial Volatility Forecasting

Which organisation /person is providing the data?	Oxford-Man Institute of Quantitative Finance, University of Oxford (realised volatility library); daily SPY and VIX market data via the yfinance interface to public exchange/market data.
Are you going to <u>only</u> use secondary data for your project? (i.e. you will not be collecting any new data)	Yes
Are you going to take (scrape / mine) data from social media feeds?	No
Are the data publicly accessible / available? (this includes datasets which are available to subscribers or can be purchased)	Yes
If the data is not publicly accessible / available, is there an agreement in place for using the data? (e.g. see agreement next page)	NA
Has the data been anonymised?	Yes
Is there no risk of re-identification of individuals? (This could be for instance by combining the data with other data on same groups, or rare, unusual or low number data. In general, this would be if re-identification is possible by "means reasonably likely to be used")	Yes
Are the data being managed and shared by reputable external organisations or institutions (that comply with all legal requirements concerning data protection)? (examples are NHS, central government, local authorities, public bodies, for example accessed via https://www.data.gov.uk or https://opendatascience.com/15-open-datasets-for-healthcare or https://www.england.nhs.uk/)	Yes
Is the proposed research in line with the website / organisation / institution sharing agreement, terms and conditions? (the terms and conditions that the organisation / person is sharing the data will vary, and may have conditions on sharing [e.g. not for commercial use, must always reference original paper/author])	Yes
Does consent exist in the original study / data collection to share the data, and is the proposed research in line with the participants original consent? (if possible, include a blank copy of the original consent form in your dissertation from original study that participants have agreed for their anonymised data to be shared for further research [or some other evidence])	N/A (no human participants; data is aggregate market data with no personal information)
Is your data management plan adequate for the data? (this will be access, storage, processing and preservation/deletion of data)	Yes
Does the organisation / person sharing the data require evidence of an ethical review for use of the data?	No

Supervisors: Please sign below to state that 1) only secondary data used for the project, 2) no data are being taken from social media feeds, 3) the data being used contain no personal data (is de-identified/ anonymous), 4) has existing consent from participants for sharing and use in line with the research, 5) is able to be used legally in the way you propose (in terms and conditions / agreement of data use), 6) has no risk of re-identification, 7) data management is adequate for the data, and 8) the organisation / person / institute managing and sharing the data complies with data processing regulations and does not require an ethical review for sharing.

Supervisor Name: Dr Tom Thorne Signature: *Tom Thorne* Date: 24/8/26

Student Name: Danish Whelan Bin Zamri Student Level (e.g. UG/ PGT): PGT